

# Day 2 – How Machine Learning Actually Learns

This document is a **complete, content-first Day 2 teaching presentation**.

It is designed to be copied directly into slides (PowerPoint / Google Slides).

No focus on visuals or design — the emphasis is on **conceptual clarity and teaching flow**.

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## Slide 1 — Title

### Day 2 — Machine Learning Intuition

How models learn and why it matters

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## Slide 2 — Recap of Day 1

From Day 1, we established: - AI is pattern learning from data, not human intelligence - LLMs are a subset of AI, not the whole story - Most AI failures happen before coding begins - AI vs ML vs Deep Learning distinctions matter

**Discussion Prompt:** What is one mental model from Day 1 that changed how you think about AI?

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## Slide 3 — Homework Review: Andrew Ng – AI for Everyone

You were asked to review the introduction from Andrew Ng's *AI for Everyone*.

Key themes participants should have captured: - AI is a broad concept; Machine Learning is a way to achieve AI - ML systems learn mappings from inputs to outputs using data - Deep Learning uses neural networks and sits under ML - AI systems do not reason like humans; they detect patterns - Understanding AI concepts is more important than knowing tools

**Instructor Prompt:** Ask 2–3 participants to summarize what stood out most to them.

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## Slide 4 — Homework Review: StatQuest – Machine Learning in 5 Minutes

From the StatQuest video, key takeaways include: - Machine Learning is about learning patterns from examples - Models generalize from known data to unseen data - ML works well when patterns repeat - Simple intuition explains most ML behavior

**Discussion Prompt:** What did this video make clearer than the Andrew Ng lecture?

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## Slide 5 — Why Today's Topic Matters

Today we focus on: - How Machine Learning actually learns - Why there are different types of learning - How this understanding helps you choose the right AI approach

This session builds the foundation for **problem identification and model selection**.

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## Slide 6 — What Is Machine Learning? (Intuitive View)

Machine Learning is: - A way for systems to learn from data - Not rules written explicitly by engineers - A process of learning input-output relationships

Instead of coding logic, we provide examples and let the model learn patterns.

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## Slide 7 — Training vs Prediction

Machine Learning has two key phases:

**Training:** - Model sees data (and often labels) - Learns patterns and relationships

**Prediction (Inference):** - Model sees new data - Produces an output based on learned patterns

This cycle is the core of all ML systems.

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## Slide 8 — Why Data Matters More Than Models

Key truths: - Good data beats complex models - Poor data guarantees poor results - Labels define what the model learns

Machine Learning success is mostly a data problem, not an algorithm problem.

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## Slide 9 — Types of Machine Learning

There are three primary learning paradigms:

1. **Supervised Learning**
2. **Unsupervised Learning**
3. **Reinforcement Learning**

Today's focus will be on supervised and unsupervised learning.

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## Slide 10 — Supervised Learning (Core Idea)

Supervised Learning: - Uses labeled data - Learns a mapping from inputs to outputs - Most common ML approach in industry

Examples: - Spam detection - Fraud detection - Price prediction

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## Slide 11 — Supervised Learning: Classification Example

Scenario: - Input: Email text - Output: Spam or Not Spam

The model learns statistical patterns linking text features to labels.

This is classification — a core ML problem type.

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## Slide 12 — Supervised Learning: Regression Example

Scenario: - Input: House features - Output: Price

Regression predicts continuous numeric values instead of categories.

This is one of the simplest and most powerful ML use cases.

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## Slide 13 — What Is Unsupervised Learning?

Unsupervised Learning: - Uses unlabeled data - Discovers structure or patterns - Does not predict predefined outputs

Common uses: - Customer segmentation - Pattern discovery

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## Slide 14 — When to Use Each Learning Type

Use **Supervised Learning** when: - You have labeled data - You want specific predictions

Use **Unsupervised Learning** when: - Labels are unavailable - You want insights or grouping

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## Slide 15 — Where Machine Learning Fits in AI

Machine Learning: - Is a major way to build AI systems - Powers many modern applications - Includes deep learning and LLMs

Understanding ML helps you reason about all AI systems.

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## Slide 16 — Reflection Exercise

Think about systems you've built: - Would any of them benefit from ML? - If yes, which type of learning? - If not, why?

This reflection builds intuition for real-world usage.

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## Slide 17 — Common Misconceptions

- More data always fixes problems (false)
- Labels are optional (sometimes false)
- ML equals deep learning (false)

Clear thinking avoids wasted AI effort.

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## Slide 18 — Interactive Assessment

For each scenario, answer: 1. Is ML appropriate? 2. Which type of learning? 3. What data is needed?

Examples: - Spam detection - House price prediction - Customer clustering - Anomaly detection

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## Slide 19 — Key Takeaways

- Machine Learning learns patterns from data
  - Training and inference are distinct phases
  - Supervised and unsupervised learning solve different problems
  - Data quality drives success
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## Slide 20 — What Comes Next

Next sessions will cover: - Classification vs Regression in depth - How to map business problems to ML tasks - Evaluating ML solutions

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## **Slide 21 — Homework**

Before the next session: - Identify 1–2 real problems from your work - Describe: - What data exists - Whether ML is applicable - Which learning type you would choose

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**End of Day 2 Teaching Presentation**